Composite Observer-Based Resilient Adaptive Fault-Tolerant Control for Interconnected Nonlinear Systems Under Deception Attacks

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***Abstract*—This article proposes a decentralized resilient adaptive intermittent output feedback fault-tolerant control (FTC) method for interconnected nonlinear systems with sensor deception attacks, multiple actuator faults, and input saturation. First, we constructed a new composite observer using intermittent compromised output signals and fuzzy logic systems (FLSs) to estimate the state variables in the face of deception attacks and unknown external disturbances. On this basis, the reasonable adaptive laws are cleverly devised to handle multiple unpredictable actuator faults. Then, in order to deal with unknown gains caused by deception attacks, a projection operator-based compensation mechanism is proposed. In addition, the "explosion of complexity" in the traditional backstepping design process is solved through command filtering technology. At the same time, an error compensation system and an auxiliary system are designed to reduce the errors caused by command filtering and solve the input saturation problem, respectively. The results indicate that the proposed control scheme can ensure the boundedness of all signals in the closed-loop system, and the tracking errors converge to a bounded compact set. Finally, the effectiveness of the proposed strategy was verified through numerical simulation.**

***Index Terms*—Actuator faults, backstepping, deception attacks, event-triggered, input saturation, interconnected nonlinear systems.**

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I. Introduction

In PRACTICAL applications, the efficient completion of tasks usually relies on the collaboration of multiple subsystems that are interconnected and interact with each other. However, due to the structural complexity and dynamic coupling of interconnected systems, centralized control approaches are often difficult to implement effectively. To address this issue, decentralized control has been proposed and has achieved a series of significant results [1], [2], [3]. Notably, due to the dependence of interconnected systems on open communication networks, these systems are vulnerable to cyber attacks including deception attacks [4], [5], [6], replay attacks [7], [8], and denial-of-service (DoS) attacks [9], [10]. Compared to other attacks, deception attacks may pose greater threats and risks to the system due to their concealment and wide range of targets. However, research on nonlinear interconnected systems subjected to deception attacks remains relatively limited. In [11], a decentralized adaptive control strategy was proposed for nonlinear interconnected systems under deception attacks, using the Nussbaum and flat-zone Lyapunov functions. Although this method achieves satisfactory performance, it relies on complex Nussbaum-type functions, which increase computational burden and complicate the controller design. Therefore, developing a decentralized control scheme with a simple structure and high computational efficiency to counter unknown deception attacks remains an important topic worthy of further investigation.

In addition, in practical applications, sensors often face limitations in energy and communication bandwidth during signal acquisition and transmission. To address this problem, the event-triggered control (ETC) strategy has been proposed [12], [13], [14]. An ETC strategy to solve the entry capture problem of uncertain nonlinear systems under asymmetric performance constraints is proposed in [12]. However, the above studies rely on the full availability of system states to design the ETC scheme, while in most practical applications, it is often difficult to obtain all the system states. Therefore, designing ETC schemes solely based on output signals has important research value [15], [16]. In [16], an event-triggered adaptive neural network control method is designed for uncertain nonlinear systems, addressing unmeasurable states and discontinuous output signals. Notably, the above research mainly focuses on general nonlinear systems, and how to develop ETC strategies

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for interconnected nonlinear systems remains a challenging problem.

On the other hand, actuator faults are usually inevitable due to factors such as harsh environments, hardware wear, and external interference. Actuator faults may lead to incorrect responses to control inputs, thereby deteriorating system performance. Thus, fault-tolerant control (FTC) has become a significant research topic in the past few decades [17], [18], [19]. In [17], an optimal active fuzzy FTC method was proposed for multiinput multioutput nonlinear systems with actuator and sensor faults, ensuring adaptive and robust stability. Input saturation is a nonlinear constraint commonly encountered in practical systems due to actuator limitations, safety requirements, and physical constraints. In fact, in many practical physical systems, input saturation and actuator fault often occur simultaneously. To address the above issues, various FTC methods have been proposed. For example, a neural network-based adaptive FTC for nonlinear systems with actuator failures and input constraints is proposed in [22]. However, the control strategy proposed in [22] is not applicable to nonlinear interconnected systems and does not consider how to ensure the stability of output-feedback systems under network attacks. Therefore, how to realize FTC for nonlinear interconnected systems under network attacks remains an open and challenging problem.

To address the aforementioned challenges, this article develops a decentralized adaptive intermittent output feedback tracking control scheme for a class of nonlinear interconnected systems subject to multiple actuator faults, input saturation, and deception attacks. The key contributions are outlined as follows.

1) In existing studies [12] and [23], the proposed ETC schemes are mainly limited to normal-form nonlinear systems. In contrast, this article investigates a more general class of interconnected nonlinear systems with mismatched uncertainties. Furthermore, unlike the methods in [24], [25], and [26], a unified control framework is adopted in this article, which does not require separate controller designs for communication scenarios with or without event-triggered mechanisms (ETMs), thereby reducing the complexity of control design.
2) Unlike [27] and [28], the original output signals in this article are rendered unreliable due to malicious deception attacks, which make conventional observer designs ineffective. To overcome this limitation, a novel composite observer is proposed based on intermittently available compromised output signals, achieving simultaneous estimation of the system states and external disturbances. Furthermore, compared with [29] and [30], this article relaxes the conventional restrictive assumption that the attack signals must be strictly positive, and proposes a new adaptive attack compensation framework utilizing projection operators to effectively address the challenges caused by unknown time-varying gains induced by deception attacks.
3) Compared with [19] and [31], the FTC method proposed in this article successfully removes the restrictive assumption of known actuator faults. By properly designing adaptive laws, a compensation framework is constructed to effectively handle unpredictable local and nonlocal actuator faults. In addition, an auxiliary system is ingeniously designed to cope with input saturation, thereby enhancing the robustness of the system and its applicability in practical engineering applications.

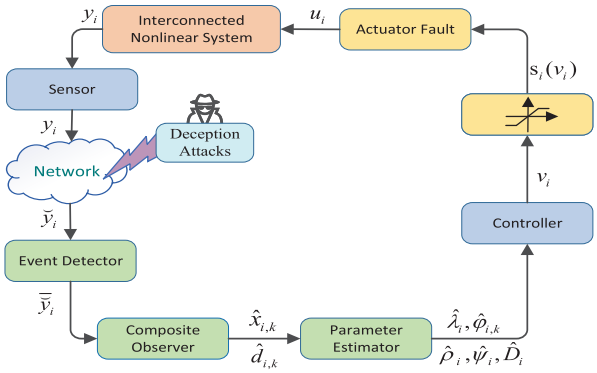


Fig. 1. Logical framework of the proposed control strategy.

II. Preliminaries and Problem Formulation

A. Problem Formulation

Consider the dynamic model of the interconnected nonlinear system as follows:

$$\begin{aligned}\dot{x}_{i,k} &= x_{i,k+1} + f_{i,k}\left(\underline{x}_{i,k}\right) + M_{i,k}\left(\underline{y}_i\right) + d_{i,k}(t)\\ \dot{x}_{i,n} &= \sum_{\mathfrak{a}=1}^{m_i} u_{i,\mathfrak{a}} + f_{i,n}\left(\underline{x}_{i,n}\right) + M_{i,n}\left(\underline{y}_i\right) + d_{i,n}(t)\\ y_i &= x_{i,1}\end{aligned} \tag{1}$$

with

$$\check{y}_i = x_{i,1}(t) + \wp_{i,1}(t) \tag{2}$$

where $i = 1,\ldots,\mathcal{N}$, $k = 1,\ldots,n-1$; $\underline{x}_{i,k} = [x_{i,1},\ldots,x_{i,k}]^T$ and $\underline{x}_{i,n} = [x_{i,1},\ldots,x_{i,n}]^T$ represent the state vectors of the ith subsystem, y_i represents the system output, $u_{i,a}$ represents the system input, m_i represents the number of actuators in the ith subsystem, $f_{i,k}(\underline{x}_{i,k})$ and $f_{i,n}(\underline{x}_{i,n})$ represent the unknown smooth nonlinear functions, $M_{i,k}(\underline{y}_i)$ and $M_{i,n}(\underline{y}_i)$ represent uncertain interactions between subsystems with $\underline{y}_i = [y_1,\ldots,y_{\mathcal{N}}]^T$, and the bounded and continuous functions $d_{i,k}(t)$ and $d_{i,n}(t)$ represent the unknown external disturbances. $\wp_{i,1}(t) = \varsigma(t)x_{i,1}(t)$ represents the malicious deception attack, where $\varsigma(t)$ is the unknown time-varying weight.

Fig. 1 shows the logical control framework of the system. In practical applications, unexpected situations such as voltage reduction, mechanical wear, and so on may cause actuators to fault during system operation. In addition, due to the physical limitations of actuators, such as the limited range of motion of robot joints, input saturation may also occur. These faults and saturation phenomena often lead to deviations between the designed control signal $v_{i,\mathfrak{a}}$ and the actual control signal $u_{i,\mathfrak{a}}$, thereby affecting the performance and stability of the system.

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The connection between $v_{i,\mathfrak{a}}$ and $u_{i,\mathfrak{a}}$ (i.e., actuator fault model) can be represented as

$$u_{i,\mathfrak{a}}(t) = \rho_{i,\mathfrak{a}} \mathrm{s}_{i,\mathfrak{a}}\left(v_{i,\mathfrak{a}}(t)\right) + u^{\star}_{i,\mathfrak{a}}(t), \quad \rho_{i,\mathfrak{a}} u^{\star}_{i,\mathfrak{a}}(t) = 0 \tag{3}$$

where $i = 1,\ldots,\mathcal{N}$, $\mathfrak{a} = 1,\ldots,m_i$, $0 \le \rho_{i,\mathfrak{a}} \le 1$ denotes the actuator's effectiveness, and $u^{\star}_{i,\mathfrak{a}}$ represents an unknown time-varying additive fault that satisfies $|u^{\star}_{i,\mathfrak{a}}(t)| \le \bar{u}^{\star}_{i,\mathfrak{a}}$. $\mathrm{s}_{i,\mathfrak{a}}(v_{i,\mathfrak{a}})$ represents the saturated control signal, as shown below

$$\mathrm{s}_{i,\mathfrak{a}}\left(v_{i,\mathfrak{a}}\right) = \begin{cases} \bar{v}_{i,\mathfrak{a}}, & \text{if } v_{i,\mathfrak{a}} > \bar{v}_{i,\mathfrak{a}} \\ v_{i,\mathfrak{a}}, & \text{if } \underline{v}_{i,\mathfrak{a}} \le v_{i,\mathfrak{a}} \le \bar{v}_{i,a} \\ \underline{v}_{i,\mathfrak{a}}, & \text{if } v_{i,\mathfrak{a}} < \underline{v}_{i,\mathfrak{a}} \end{cases} \tag{4}$$

with $\bar{v}_{i,\mathfrak{a}}$ and $\underline{v}_{i,\mathfrak{a}}$ denote the upper and lower bounds of the ath actuator in the ith subsystem, respectively.

Remark 1: According to (3), it can be inferred that if $\rho_{i,\mathfrak{a}} = 1$ and $u^{\star}_{i,\mathfrak{a}} = 0$, then the actuator is not faulty; if $0 < \rho_{i,\mathfrak{a}} < 1$ and $u^{\star}_{i,\mathfrak{a}} = 0$, then the actuator only has $\rho_{i,\mathfrak{a}}$ effectiveness; if $\rho_{i,\mathfrak{a}} = 0$, then the actuator loses all effectiveness, and when $u^{\star}_{i,\mathfrak{a}} \neq 0$, it will suffer from bias fault. To guarantee system controllability, we assume that at most $m_i - 1$ actuators completely lose effectiveness simultaneously, i.e., $\sum_{\mathfrak{a}=1}^{m_i} \rho_{i,\mathfrak{a}} > 0$.

Assumption 1: $y_{i,d}$ is the n-order differentiable desired trajectory of the ith subsystem, satisfying $|y_{i,d}| \le Y_{i,d}$ for a constant $Y_{i,d} > 0$.

Assumption 2: The continuous time-varying disturbance $d_{i,k}(t)$ is bounded, and there exists a constant $d_{i,M} > 0$ such that $|\dot{d}_{i,k}(t)| \le d_{i,M}$.

Assumption 3: The time-varying weight $\varsigma(t)$ satisfies $1 + \varsigma(t) \neq 0$, and there exist positive constants $\bar{\varsigma}$ and $\bar{\dot{\varsigma}}$ such that $|\varsigma| \le \bar{\varsigma}$ and $|\dot{\varsigma}| \le \bar{\dot{\varsigma}}$. Further define $\lambda_i(t) = (1 + \varsigma(t))^{-1}$, it can be known that there are positive constants $\lambda_{i,M}$ and $\bar{\dot{\lambda}}_i$ such that $|\lambda_i| \le \lambda_{i,M}$ and $|\dot{\lambda}_i| \le \bar{\dot{\lambda}}_i$.

Assumption 4: There exists a known smooth positive function $h_{ik\mathfrak{q}}(\cdot)$ such that

$$M^2_{i,k}\left(\underline{y}_i\right) \le \sum_{\mathfrak{q}=1}^{\mathcal{N}} h_{ik\mathfrak{q}}\left(x_{\mathfrak{q},1}\right) + \delta_{i,k} \tag{5}$$

where $h_{ik\mathfrak{q}}(0) = 0$ and $\delta_{i,k}$ is a positive constant.

Remark 2: According to [32], it can be known that Assumption 4 is reasonable. Since $h_{ik\mathfrak{q}}(0) = 0$, there exists a smooth function $\hbar_{ik\mathfrak{q}}(x_{\mathfrak{q},1})$ that makes $h_{ik\mathfrak{q}}(x_{\mathfrak{q},1}) = x_{\mathfrak{q},1}\hbar_{ik\mathfrak{q}}(x_{\mathfrak{q},1})$, further assuming that for any $\chi_1, \chi_2 \in R$, there exists a positive constant $l_{ik\mathfrak{q}}$ that makes $|\hbar_{ik\mathfrak{q}}(\chi_1) - \hbar_{ik\mathfrak{q}}(\chi_2)| \le l_{ik\mathfrak{q}}|\chi_1 - \chi_2|$.

B. Event-Triggered Mechanism

For the purpose of saving communication resources, the ETM is defined as

$$\begin{aligned} &\bar{\check{y}}_i(t) = \check{y}_i\left(t_{i,\mathfrak{p}}\right), \quad t \in \left(t_{i,\mathfrak{p}}, t_{i,\mathfrak{p}+1}\right) \\ &t_{i,\mathfrak{p}+1} = \inf\left\{t > t_{i,\mathfrak{p}}, |\check{y}_i(t) - \bar{\check{y}}_i(t)| \ge \gamma_i|\check{y}_i| + \ell_i\right\} \end{aligned} \tag{6}$$

where γ_i and ℓ_i are positive design constants, $t_{i,\mathfrak{p}}(\mathfrak{p} = 0, 1, 2, \ldots,)$ represents the $\mathfrak{p}$th event-triggered instant of the compromised output signal $\check{y}_i$ with $t_{i,0}$ is the first triggering moment. When trigger condition (6) is met, the triggered compromised output signal $\bar{\check{y}}_i$ will update and remain unchanged until the next event triggering occurs.

C. Fuzzy Logic Systems

In order to handle the unknown nonlinear smooth functions present in the interconnected system (1), fuzzy logic systems (FLSs) are utilized to approximate them over a suitable compact set. By adopting the singleton fuzzifier, the product inference mechanism, and the center-average defuzzifier, the fuzzy rules are constructed as follows:

R^i: if ζ_1 is C^i_1 and ... and ζ_n is C^i_n then y is B^i

where $\boldsymbol{\zeta} = [\zeta_1, \ldots, \zeta_n]^T$ denotes the input and y denotes the output. $C^i_1, \ldots, C^i_n$ and B^i are fuzzy sets in $\mathbb{R}$. Following [27], the output of the FLSs can be written as

$$y(\boldsymbol{\zeta}) = \frac{\sum_{i=1}^{\mathcal{J}} \varphi^*_i \prod_{b=1}^{n} \aleph_{C^i_b}(\zeta_b)}{\sum_{i=1}^{\mathcal{J}} \left[\prod_{b=1}^{n} \aleph_{C^i_b}(\zeta_b)\right]}$$

where φ^*_i represents the point for which the fuzzy membership function $\aleph_{C^i}(\varphi^*_i) = 1$. Let $\boldsymbol{Q}(\boldsymbol{\zeta}) = \left(\prod_{b=1}^{n} \aleph_{C^i_b}(\zeta_b)\right) / \left(\sum_{i=1}^{\mathcal{J}} \left[\prod_{b=1}^{n} \aleph_{C^i_b}(\zeta_b)\right]\right)$, $\boldsymbol{Q}(\boldsymbol{\zeta}) = [Q_1(\boldsymbol{\zeta}), \ldots, Q_{\mathcal{J}}(\boldsymbol{\zeta})]^T$ denotes the basic function vector, $\boldsymbol{\varphi}^* = [\varphi^*_1, \ldots, \varphi^*_{\mathcal{J}}]^T$ denotes the fuzzy weighting vector, the FLSs can be rephrased as

$$y(\boldsymbol{\zeta}) = \boldsymbol{\varphi}^{*T} \boldsymbol{Q}(\boldsymbol{\zeta}). \tag{7}$$

where $0 < \boldsymbol{Q}^T(\boldsymbol{\zeta})\boldsymbol{Q}(\boldsymbol{\zeta}) \le 1$.

D. Projection Operator-Based Deception Attack Compensation

The continuously differentiable convex function $\Phi(\hat{\lambda}_i) = ((\mathfrak{c}_i + 1)\hat{\lambda}^T_i\hat{\lambda}_i - \hat{\lambda}^2_{i,M})/(\mathfrak{c}_i\hat{\lambda}^2_{i,M})$, where $\mathfrak{c}_i > 0$ is a projection tolerance bound, $\hat{\lambda}_i$ is the deception attack compensator signal, and $\hat{\lambda}_{i,M}$ is the projection norm bound of $\hat{\lambda}_i$, we can get

$$\dot{\hat{\lambda}}_i = \begin{cases} \beth_i, & \text{if } \Phi\left(\hat{\lambda}_i\right) < 0 \text{ or } \Phi\left(\hat{\lambda}_i\right) \\ & \quad \ge 0 \text{ and } \boldsymbol{\Phi}'\left(\hat{\lambda}_i\right)\beth_i \le 0 \\ P_i[\beth_i], & \text{if } \Phi\left(\hat{\lambda}_i\right) \ge 0 \text{ and } \boldsymbol{\Phi}'\left(\hat{\lambda}_i\right)\beth_i > 0 \end{cases} \tag{8}$$

where $\beth_i = (q_{i,1})/2\left(\|\mathcal{P}_i\|^2\|\|\mathcal{L}_i\|^2 + \sum_{s=1}^{n} l^2_{i,s}\right)\bar{\bar{x}}^2_{i,1} - q_{i,1}b_i\hat{\lambda}_i$.

The projection operator is

$$P_i[\beth_i] = \beth_i - \frac{\boldsymbol{\Phi}'\left(\hat{\lambda}_i\right)\boldsymbol{\Phi}'^T\left(\hat{\lambda}_i\right)}{\left\|\boldsymbol{\Phi}'\left(\hat{\lambda}_i\right)\right\|^2}\Phi\left(\hat{\lambda}_i\right)\beth_i. \tag{9}$$

E. Necessary Lemmas

Lemma 1 [27]: For continuous function $F(\boldsymbol{\zeta})$ on the compact set Ω, we can obtain

$$\sup_{\boldsymbol{\zeta}\in\Omega}\left|F(\boldsymbol{\zeta}) - \boldsymbol{\varphi}^{*T}Q(\boldsymbol{\zeta})\right| \le \varrho. \tag{10}$$

where constant $\varrho > 0$.

Lemma 2 [33]: For all $\eth_1 \in \mathbb{R}$ and $\eth_2 \in \mathbb{R}$, the inequality below is valid

$$|\eth_1|^{\mathfrak{o}_1}|\eth_2|^{\mathfrak{o}_2} \le \frac{\mathfrak{o}_1\mathfrak{o}_3}{\mathfrak{o}_1 + \mathfrak{o}_2}|\eth_1|^{\mathfrak{o}_1+\mathfrak{o}_2} + \frac{\mathfrak{o}_2\mathfrak{o}_3^{-\frac{\mathfrak{o}_1}{\mathfrak{o}_2}}}{\mathfrak{o}_1 + \mathfrak{o}_2}|\eth_2|^{\mathfrak{o}_1+\mathfrak{o}_2} \tag{11}$$

where $\mathfrak{o}_1$, $\mathfrak{o}_2$, and $\mathfrak{o}_3$ are positive constants.

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Lemma 3 [34]: For all variable $\Upsilon \in \mathbb{R}$ and constant $\varepsilon > 0$, the hyperbolic tangent function exhibits the following characteristics:

$$\Upsilon \tanh\left(\frac{\Upsilon}{\varepsilon}\right) \geq 0$$
$$0 \leq |\Upsilon| - \Upsilon \tanh\left(\frac{\Upsilon}{\varepsilon}\right) \leq 0.2785\varepsilon. \tag{12}$$

Lemma 4 [35]: For any variable $K \in \mathbb{R}$ and constant $\epsilon > 0$, $\lim_{K\to 0} \tanh^2(K/\epsilon)/K = 0$. For the set $\Omega_\epsilon = \{K \| K| \leq 0.8814\epsilon\}$, the inequality $1 - 2\tanh^2(K/\epsilon) \leq 0$ holds for $K \notin \Omega_\epsilon$.

Lemma 5 [36]: For any $\iota > 0$ and $\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_q \in \mathbb{R}$, one has $(\mathcal{X}_1 + \cdots + \mathcal{X}_q)^\iota \leq \max\{q^{\iota-1}, 1\}(\mathcal{X}_1^\iota + \cdots + \mathcal{X}_q^\iota)$.

III. Main Results

A. Composite Observer Design

Regarding the unknown function $f_{i,k}(\underline{\boldsymbol{x}}_{i,k})$ and uncontrollable additive fault $u_{i,\mathfrak{a}}^\star$ in system (1), this article uses FLSs and Lemma 1 to rephrase system (1) as follows:

$$\begin{aligned}
\dot{x}_{i,k} &= x_{i,k+1} + \varphi_{i,k}^{*T} Q_{i,k}\left(\underline{\boldsymbol{x}}_{i,k}\right) + M_{i,k}\left(\underline{\boldsymbol{y}}_i\right) + d_{i,k}(t) + \varrho_{i,k} \\
\dot{x}_{i,n} &= \boldsymbol{\rho}_i^T \mathbf{s}_i(\boldsymbol{v}_i) + \varphi_{i,n}^{*T} Q_{i,n}\left(\underline{\boldsymbol{x}}_{i,n}\right) \\
&\quad + M_{i,n}\left(\underline{\boldsymbol{y}}_i\right) + d_{i,n}(t) + \varrho_{i,n} \\
y_i &= x_{i,1}, \quad k = 1, \ldots, n-1
\end{aligned} \tag{13}$$

where $|f_{i,k}(\underline{\boldsymbol{x}}_{i,k}) - \varphi_{i,k}^{*T} Q_{i,k}(\underline{\boldsymbol{x}}_{i,k})| \leq \varrho_{i,k}$, $\varphi_{i,k}^{*T} Q_{i,k}(\underline{\boldsymbol{x}}_{i,k})$ represents the approximations of unknown function, $\varrho_{i,k}$ represents the approximation error that satisfy $|\varrho_{i,k}| \leq \bar{\varrho}_{i,k}$, with $\bar{\varrho}_{i,k}$ being positive constant, for $k = 1, \ldots, n$. $\boldsymbol{\rho}_i = [\rho_{i,1}, \ldots, \rho_{i,m_i}]^T$, and $\mathbf{s}_i(\boldsymbol{v}_i) = [\mathrm{s}_{i,1}(v_{i,1}), \ldots, \mathrm{s}_{i,m_i}(v_{i,m_i})]^T$.

In this article, due to a malicious deception attack on the system, the output signal $x_{i,1}$ cannot be measured and used, so only the compromised state $\check{x}_{i,1}$ can be used for control design. Furthermore, to reduce communication and computational burdens, this article considers a more challenging problem, in which only the triggered compromised state $\bar{\check{x}}_{i,1}$ is available for control design rather than the compromised state $\check{x}_{i,1}$. To estimate the unknown states $x_{i,k}(k = 2, \ldots, n)$ and handle external disturbance d_i, a novel fuzzy composite observer is constructed, as shown below

$$\begin{aligned}
\dot{\hat{x}}_{i,k} &= \hat{x}_{i,k+1} + l_{i,k}\left(\hat{\lambda}_i \bar{\check{x}}_{i,1} - \hat{x}_{i,1}\right) + \hat{\varphi}_{i,k}^T Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right) + \hat{d}_{i,k} \\
\dot{\hat{x}}_{i,n} &= \hat{\boldsymbol{\rho}}_i^T \mathbf{s}(\boldsymbol{v}_i) + l_{i,n}\left(\hat{\lambda}_i \bar{\check{x}}_{i,1} - \hat{x}_{i,1} + \hat{\varphi}_{i,n}^T Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) + \hat{d}_{i,n}\right) \\
\hat{y}_i &= \hat{x}_{i,1}, \quad k = 1, \ldots, n-1
\end{aligned} \tag{14}$$

where $l_{i,k}$ is the designed parameter, $\hat{x}_{i,k}$ represents the estimate of $x_{i,k}$, $\hat{\varphi}_{i,n}^T$ represents the estimate of $\varphi_{i,n}^{*T}$, $\tilde{\varphi}_{i,n}^T = \varphi_{i,n}^{*T} - \hat{\varphi}_{i,n}^{*T}$, and $\hat{d}_{i,k}$ represents the estimate of $d_{i,k}$, $\tilde{d}_{i,k} = d_{i,k} - \hat{d}_{i,k}$, for $k = 1, \ldots, n$. $\hat{\lambda}_i$ represents the estimate of λ_i, $\tilde{\lambda}_i = \lambda_i - \hat{\lambda}_i$, and $\hat{\boldsymbol{\rho}}_i$ represents the estimate of $\boldsymbol{\rho}_i$, $\tilde{\boldsymbol{\rho}}_i = \boldsymbol{\rho}_i - \hat{\boldsymbol{\rho}}_i$.

The estimation error is represented by

$$\mathbf{e}_i = \underline{\boldsymbol{x}}_i - \underline{\hat{\boldsymbol{x}}}_i \tag{15}$$

where $\mathbf{e}_i = [\mathrm{e}_{i,1}, \ldots, \mathrm{e}_{i,n}]^T$, $\underline{\boldsymbol{x}}_i = [x_{i,1}, \ldots, x_{i,n}]^T$, and $\underline{\hat{\boldsymbol{x}}}_i = [\hat{x}_{i,1}, \ldots, \hat{x}_{i,n}]^T$.

In accordance with (13) and (14), we can get

$$\begin{aligned}
\dot{\mathbf{e}}_i &= \mathcal{A}_i \mathbf{e}_i + \sum_{k=1}^{n} \mathcal{I}_{i,k} \tilde{\varphi}_{i,k}^T Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right) \\
&\quad + \sum_{k=1}^{n} \mathcal{I}_{i,k} \varphi_{i,k}^T \left(Q_{i,k}\left(\underline{\boldsymbol{x}}_{i,k}\right) - Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right)\right) \\
&\quad + \boldsymbol{\mathcal{L}}_i \tilde{\lambda}_i \bar{\check{x}}_{i,1} + \boldsymbol{\mathcal{L}}_i \lambda_i \left(\bar{\check{x}}_{i,1} - \bar{\check{x}}_{i,1}\right) \\
&\quad + \mathcal{I}_{i,n} \tilde{\boldsymbol{\rho}}_i^T \mathbf{s}_i(\boldsymbol{v}_i) + \boldsymbol{\varrho}_i + \tilde{\boldsymbol{d}}_i + \boldsymbol{M}_i\left(\underline{\boldsymbol{y}}_i\right)
\end{aligned} \tag{16}$$

where

$$\mathcal{A}_i = \begin{bmatrix} -l_{i,1} & & \\ \vdots & & I_{n-1} \\ -l_{i,n} & \cdots & 0 \end{bmatrix}$$

$\boldsymbol{\mathcal{L}}_i = [l_{i,1}, \ldots, l_{i,n}]^T$, $\tilde{\boldsymbol{d}}_i = [\tilde{d}_{i,1}, \ldots, \tilde{d}_{i,n}]^T$, $\boldsymbol{\varrho}_i = [\varrho_{i,1}, \ldots, \varrho_{i,n}]^T$, $\boldsymbol{M}_i(\underline{\boldsymbol{y}}_i) = [M_{i,1}(\underline{\boldsymbol{y}}_i), \ldots, M_{i,n}(\underline{\boldsymbol{y}}_i)]^T$,

$$\mathcal{I}_{i,k} = \left[\underbrace{0, \cdots, 0}_{k-1}, 1, 0, \ldots, 0\right]^T$$

and

$$\mathcal{I}_{i,n} = \left[\underbrace{0, \cdots, 0}_{n-1}, 1\right]^T.$$

Choosing the appropriate vector $\boldsymbol{\mathcal{L}}_i$ can make $\mathcal{A}_i$ a Hurwitz matrix, so there exists $\mathcal{P}_i = \mathcal{P}_i^T > 0$, which holds $\mathcal{A}_i^T \mathcal{P}_i + \mathcal{P}_i \mathcal{A}_i = -2\mathfrak{Q}_i$, where $\mathfrak{Q}_i$ is a positive definite matrix.

To address the effects caused by the unknown disturbance d_i on the system, an intermediate variable W_i is constructed as shown below

$$\boldsymbol{W}_i = \boldsymbol{d}_i - \pi_i \underline{\boldsymbol{x}}_i, i = 1, \ldots, \mathcal{N} \tag{17}$$

where $W_i = [W_{i,1}, \ldots, W_{i,n}]^T$ and $\pi_i = \mathrm{diag}(\pi_{i,1}, \ldots, \pi_{i,n}) \in R^{n\times n}$ with $\pi_{i,k} > 0$ is constant for $k = 1, \ldots, n$.

By differentiating (17) and using (1), we obtain

$$\begin{aligned}
\dot{W}_{i,k} &= \dot{d}_{i,k} - \pi_{i,k}\left[x_{i,k+1} + \varphi_{i,k}^{*T} Q_{i,k}\left(\underline{\boldsymbol{x}}_{i,k}\right) + \varrho_{i,k}\right. \\
&\qquad \left. + M_{i,k}\left(\underline{\boldsymbol{y}}_i\right) + W_{i,k} + \pi_{i,k} x_{i,k}\right] \\
\dot{W}_{i,n} &= \dot{d}_{i,n} - \pi_{i,n}\left[\boldsymbol{\rho}_i^T \mathbf{s}_i(\boldsymbol{v}_i) + \varphi_{i,n}^{*T} Q_{i,n}\left(\underline{\boldsymbol{x}}_{i,n}\right) + \varrho_{i,n}\right. \\
&\qquad \left. + M_{i,n}\left(\underline{\boldsymbol{y}}_i\right) + W_{i,n} + \pi_{i,n} x_{i,n}\right].
\end{aligned} \tag{18}$$

The estimation design of auxiliary variable W_i is given below

$$\begin{aligned}
\dot{\hat{W}}_{i,k} &= -\pi_{i,k}\left[\hat{x}_{i,k+1} + \hat{\varphi}_{i,k}^{*T} Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right) + \hat{W}_{i,k} + \pi_{i,k}\hat{x}_{i,k}\right] \\
\dot{\hat{W}}_{i,n} &= -\pi_{i,n}\left[\hat{\boldsymbol{\rho}}_i^T \mathbf{s}_i(\boldsymbol{v}_i) + \hat{\varphi}_{i,n}^{*T} Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) + \hat{W}_{i,n} + \pi_{i,n}\hat{x}_{i,n}\right].
\end{aligned} \tag{19}$$

Furthermore, based on (17), the estimation of the external disturbance $\boldsymbol{d}_i$ can be represented as

$$\hat{\boldsymbol{d}}_i = \hat{\boldsymbol{W}}_i + \pi_i \underline{\hat{\boldsymbol{x}}}_i. \tag{20}$$

Let $\tilde{\boldsymbol{W}}_i = \boldsymbol{W}_i - \hat{\boldsymbol{W}}_i$, and using (18) and (19), we get

$$\begin{aligned}
\dot{\tilde{W}}_{i,k} &= \dot{d}_{i,k} - \pi_{i,k}\left[\mathrm{e}_{i,k+1} + \varphi_{i,k}^T\left(Q_{i,k}\left(\underline{\boldsymbol{x}}_{i,k}\right) - Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right)\right)\right. \\
&\qquad \left. + \tilde{\varphi}_{i,k}^T Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right) + \varrho_{i,k} + M_{i,k}\left(\underline{\boldsymbol{y}}_i\right)\right.
\end{aligned}$$

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$$+\tilde{W}_{i,k}+\pi_{i,k}\mathrm{e}_{i,k}\big]$$
$$\dot{\tilde{W}}_{i,n}=\dot{d}_{i,n}-\pi_{i,n}\big[\tilde{\boldsymbol{\rho}}_i^T\mathbf{s}_i(\boldsymbol{v}_i)+\varphi_{i,n}^T\left(Q_{i,n}\left(\underline{\boldsymbol{x}}_{i,n}\right)-Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right)\right)+\tilde{\varphi}_{i,n}^TQ_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right)+\varrho_{i,n}+M_{i,n}\left(\underline{\boldsymbol{y}}_i\right)+\tilde{W}_{i,n}+\pi_{i,n}\mathrm{e}_{i,n}\big]. \tag{21}$$

To verify the stability of the designed composite observer, the Lyapunov function $V_{i,0}$ is constructed as follows:

$$V_{i,0}=\frac{1}{2}\mathbf{e}_i^T\mathcal{P}_i\mathbf{e}_i+\frac{1}{2}\sum_{\mathfrak{s}=1}^{n}\tilde{W}_{i,\mathfrak{s}}^2. \tag{22}$$

From Young's inequality, we obtain

$$\dot{V}_{i,0}\le-\nu_{i,0}\|\mathbf{e}_i\|^2-\sum_{\mathfrak{s}=1}^{n}a_{i,0}\tilde{W}_{i,\mathfrak{s}}^2+\sum_{\mathfrak{s}=1}^{n}\left\|\tilde{\varphi}_{i,\mathfrak{s}}\right\|^2+\frac{1}{2}\|\mathcal{P}_i\|^2\left\|\mathcal{L}_i\right\|^2\bar{\lambda}_i^2\bar{\bar{x}}_{i,1}^2+\sum_{\mathfrak{s}=1}^{n}\sum_{\mathfrak{q}=1}^{\mathcal{N}}x_{\mathfrak{q},1}\hbar_{i\mathfrak{s}\mathfrak{q}}\left(\hat{x}_{\mathfrak{q},1}\right)+\frac{n}{2}\sum_{\mathfrak{q}=1}^{\mathcal{N}}x_{\mathfrak{q},1}^2+\frac{\gamma_i^2}{2}\|y_i\|^2+\left\|\tilde{\boldsymbol{\rho}}_i\right\|^2\|\check{\boldsymbol{\nu}}_i\|^2+\Gamma_{i,0} \tag{23}$$

where $\check{\boldsymbol{\nu}}_i=[\check{\nu}_{i,1},\check{\nu}_{i,2},\ldots,\check{\nu}_{i,m_i}]^T$ with $\check{\nu}_{i,a}=\max\{\bar{\nu}_{i,a},\underline{\nu}_{i,a}\}$ for $a=1,\ldots,m_i$, $\nu_{i,0}=\lambda_{min}(\Omega_i)-2-2\|\mathcal{P}\|^2-\|\mathcal{P}_i\|^2\|\pi_j\|^2-1/2\sum_{\mathfrak{s}=1}^{n}\pi_i^2-1/2\sum_{\mathfrak{s}=1}^{n}\pi_i^4-\|\mathcal{P}_i\|^2\|\mathcal{L}_i\|^2-\sum_{\mathfrak{s}=1}^{n}\sum_{q=1}^{\mathcal{N}}l_{i\mathfrak{s}q}^2$, $a_{i,0}=\pi_{i,\mathfrak{s}}-5/2\pi_{i,\mathfrak{s}}^2-2$, $\Gamma_{i,0}=\sum_{\mathfrak{s}=1}^{n}(1/2\|\mathcal{P}_i\|^2+1)\varphi_{i,\mathfrak{s}}^{*2}+\sum_{\mathfrak{s}=1}^{n}(1/2\|\mathcal{P}_i\|^2+1/2)\|\bar{o}_{i,\mathfrak{s}}\|^2+1/2\sum_{\mathfrak{s}=1}^{n}\|d_{i,M}\|^2+\sum_{\mathfrak{s}=1}^{n}\delta_{i,\mathfrak{s}}+1/2\lambda_{i,M}^2\ell_i^2$.

B. Resilient Event-Triggered Controller Design

The decentralized resilient adaptive ETC strategy is designed for system (1) using the backstepping method. First, to reduce the effect of actuator saturation on the system, an auxiliary system is designed as

$$\begin{cases}\dot{\xi}_{i,k}=\xi_{i,k+1}-\eta_{i,k}\xi_{i,k}\\ \dot{\xi}_{i,n}=\hat{\boldsymbol{\rho}}_i^T\Delta\boldsymbol{u}_i-\eta_{i,n}\xi_{i,n}\end{cases} \tag{24}$$

where $k=1,\ldots,n-1$, $\xi_{i,k}$ and $\xi_{i,n}$ are auxiliary variables, $\Delta\boldsymbol{u}_i=\mathbf{s}_i(\boldsymbol{v}_i)-\boldsymbol{v}_i$, and $\eta_{i,1}>1/2$ and $\eta_{i,k}>1(k=2,\ldots,n)$ are design parameters.

Then, the coordinate transformations are designed as

$$\begin{aligned}z_{i,1}&=\hat{x}_{i,1}-y_{i,d}-\xi_{i,1}\\ z_{i,k}&=\hat{x}_{i,k}-\theta_{i,kf}-\xi_{i,k}\end{aligned} \tag{25}$$

where $k=2,\ldots,n$, $z_{i,1}$ and $z_{i,k}$ represent the system errors. Define command filters as follows:

$$\tau_{i,k}\dot{\theta}_{i,kf}+\theta_{i,kf}=\alpha_{i,k-1},\quad \theta_{i,kf}(0)=\alpha_{i,k-1}(0) \tag{26}$$

where $\tau_{i,k}>0$ is a constant. $\alpha_{i,k}$ is the input of the filter, which is also the virtual control law that will be designed later. $\theta_{i,kf}$ represents the output of the filter. The compensated tracking error is described by

$$H_{i,k}=z_{i,k}-o_{i,k},\quad k=1,\ldots,n. \tag{27}$$

To mitigate the errors due to command filtering, the following error compensation system is proposed:

$$\begin{aligned}\dot{o}_{i,1}&=-c_{i,1}o_{i,1}-4o_{i,1}+o_{i,2}+\left(\theta_{i,2f}-\alpha_{i,1}\right)\\ \dot{o}_{i,k}&=-c_{i,k}o_{i,k}-\frac{9}{2}o_{i,k}+o_{i,k+1}+\left(\theta_{i,(k+1)f}-\alpha_{i,k}\right)\\ \dot{o}_{i,n}&=-c_{i,n}o_{i,n}-\frac{7}{2}o_{i,n}\end{aligned} \tag{28}$$

where $k=2,\ldots,n-1$, $c_{i,1}$, $c_{i,k}$ and $c_{i,n}$ are positive design parameters, $o_{i,k}(0)=0$.

From Lemma 3 in [37], one obtains that $|\theta_{i,(k+1)f}-\alpha_{i,k}|\le\mu$ and $\|o_{i,k}\|$ is bounded. If $t\to\infty$, we have

$$\lim_{t\to\infty}\left\|o_{i,k}\right\|\le\frac{\mu_i}{2\mathfrak{r}_i} \tag{29}$$

where $k=1,\ldots,n$, $\mathfrak{r}_i=1/2\min(c_{i,k})$, $\mu_i>0$ is a constant, and $o_{i,k}(0)=0$.

To obtain the main results, the specific design procedure is detailed below.

Step 1: Based on (14), (24), (25), and (27), one can derive the following equation:

$$\begin{aligned}\dot{H}_{i,1}&=\dot{\hat{x}}_{i,1}-\dot{y}_{i,d}-\dot{\xi}_{i,1}-\dot{o}_{i,1}\\ &=z_{i,2}+\theta_{i,2f}+l_{i,1}\left(\hat{\lambda}_i\bar{\bar{x}}_{i,1}-\hat{x}_{i,1}\right)+\hat{\varphi}_{i,1}^TQ_{i,1}\left(\hat{x}_{i,1}\right)+\hat{d}_{i,1}-\dot{y}_{i,d}+\eta_{i,1}\xi_{i,1}-\dot{o}_{i,1}.\end{aligned} \tag{30}$$

Define the Lyapunov function $V_{i,1}$ as

$$V_{i,1}=V_{i,0}+\frac{1}{2}H_{i,1}^2+\frac{1}{2p_{i,1}}\tilde{\varphi}_{i,1}^T\tilde{\varphi}_{i,1}+\frac{1}{2q_{i,1}}\left|\tilde{\lambda}_i\right|^3 \tag{31}$$

where $p_{i,1}>0$ and $q_{i,1}>0$ are constants.

Using (30) and (31), one gets

$$\begin{aligned}\dot{V}_{i,1}\le&\dot{V}_{i,0}+H_{i,1}\left(\alpha_{i,1}+\hat{d}_{i,1}-\dot{y}_{i,d}+\eta_{i,1}\xi_{i,1}+c_{i,1}o_{i,1}+4o_{i,1}\right)+\frac{1}{2}H_{i,2}^2+4H_{i,1}^2+\left\|\tilde{\varphi}_{i,1}\right\|^2+\frac{1}{2}l_{i,1}^2\bar{\lambda}_i^2\bar{\bar{x}}_{i,1}^2\\ &+\frac{1}{p_{i,1}}\tilde{\varphi}_{i,1}\left(p_{i,1}H_{i,1}Q_{i,1}\left(\hat{x}_{i,1}\right)-\dot{\hat{\varphi}}_{i,1}\right)+\frac{1}{2}l_{i,1}^2\left\|\mathrm{e}_{i,1}\right\|^2+\frac{\tilde{\lambda}_i^2}{q_{i,1}}\mathrm{sign}\left(\tilde{\lambda}_i\right)\left(\lambda_i-\dot{\hat{\lambda}}_i\right)+\frac{l_{i,1}^2\gamma_i^2}{2}\|y_i\|^2+\Gamma_{i,1}\end{aligned} \tag{32}$$

where $\Gamma_{i,1}=1/2\|\varphi_{i,1}^*\|^2+1/2l_{i,1}^2\lambda_{i,M}^2\ell_i^2$.

Based on (32), the virtual controller $\alpha_{i,1}$ and the adaptive law $\dot{\hat{\varphi}}_{i,1}$ can be designed as

$$\alpha_{i,1}=-c_{i,1}z_{i,1}-4z_{i,1}-\hat{d}_{i,1}+\dot{y}_{i,d}-\eta_{i,1}\xi_{i,1} \tag{33}$$
$$\dot{\hat{\varphi}}_{i,1}=p_{i,1}H_{i,1}Q_{i,1}\left(\hat{x}_{i,1}\right)-\sigma_{i,1}\hat{\varphi}_{i,1}. \tag{34}$$

Substituting (33) and (34) into (32) gives

$$\begin{aligned}\dot{V}_{i,1}\le&\dot{V}_{i,0}-c_{i,1}H_{i,1}^2+\frac{1}{2}H_{i,2}^2+\left\|\tilde{\varphi}_{i,1}\right\|^2+\frac{1}{2}l_{i,1}^2\bar{\lambda}_i^2\bar{\bar{x}}_{i,1}^2+\frac{\sigma_{i,1}}{p_{i,1}}\tilde{\varphi}_{i,1}^T\hat{\varphi}_{i,1}+\frac{1}{2}l_{i,1}^2\left\|\mathrm{e}_{i,1}\right\|^2+\frac{l_{i,1}^2\gamma_i^2}{2}\|y_i\|^2\\ &+\frac{\tilde{\lambda}_i^2}{q_{i,1}}\mathrm{sign}\left(\tilde{\lambda}_i\right)\left(\lambda_i-\dot{\hat{\lambda}}_i\right)+\Gamma_{i,1}.\end{aligned} \tag{35}$$

Step k ($2\le k\le n-1$): From (14), (24), (25), and (27), we have

$$\begin{aligned}\dot{H}_{i,k}&=\dot{\hat{x}}_{i,k}-\dot{\theta}_{i,kf}-\dot{\xi}_{i,k}-\dot{o}_{i,k}\\ &=z_{i,k}+\theta_{i,(k+1)f}+l_{i,k}\left(\hat{\lambda}_i\bar{\bar{x}}_{i,1}-\hat{x}_{i,1}\right)+\hat{\varphi}_{i,k}^TQ_{i,k}\left(\hat{x}_{i,k}'\right)+\hat{d}_{i,k}-\dot{\theta}_{i,kf}+\eta_{i,k}\xi_{i,k}-\dot{o}_{i,k}.\end{aligned} \tag{36}$$

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Then, define

$$V_{i,k} = V_{i,k-1} + \frac{1}{2}H_{i,k}^2 + \frac{1}{2p_{i,k}}\tilde{\varphi}_{i,k}^T\tilde{\varphi}_{i,k} \tag{37}$$

where $p_{i,k}$ are positive constant.

The virtual controller $\alpha_{i,k}$ and the adaptive law $\dot{\hat{\varphi}}_{i,k}$ can be designed as

$$\alpha_{i,k} = -c_{i,k}z_{i,k} - \frac{9}{2}z_{i,k} - \hat{d}_{i,k} + \dot{\theta}_{i,kf} - \eta_{i,k}\xi_{i,k} \tag{38}$$

$$\dot{\hat{\varphi}}_{i,k} = p_{i,k}H_{i,k}Q_{i,k}\left(\underline{\hat{\boldsymbol{x}}}_{i,k}\right) - \sigma_{i,k}\hat{\varphi}_{i,k}. \tag{39}$$

Substituting (38) and (39) into (37) gives

$$\begin{aligned}\dot{V}_{i,k} \le \dot{V}_{i,0} &- \sum_{\mathfrak{s}=1}^{k} c_{i,\mathfrak{s}}H_{i,\mathfrak{s}}^2 + \sum_{\mathfrak{s}=1}^{k}\left\|\tilde{\varphi}_{i,\mathfrak{s}}\right\|^2 + \sum_{\mathfrak{s}=1}^{k}\frac{1}{2}l_{i,\mathfrak{s}}^2\tilde{\lambda}_i^2\bar{\bar{x}}_{i,1}^2 \\ &+ \sum_{\mathfrak{s}=1}^{k}\frac{\sigma_{i,\mathfrak{s}}}{p_{i,\mathfrak{s}}}\tilde{\varphi}_{i,\mathfrak{s}}^T\hat{\varphi}_{i,\mathfrak{s}} + \sum_{\mathfrak{s}=1}^{k}\frac{1}{2}l_{i,\mathfrak{s}}^2\|\mathrm{e}_{i,1}\|^2 + \frac{1}{2}H_{i,k+1}^2 \\ &+ \frac{\tilde{\lambda}_i^2}{q_{i,1}}\mathrm{sign}\left(\tilde{\lambda}_i\right)\left(\lambda_i - \dot{\hat{\lambda}}_i\right) + \sum_{\mathfrak{s}=1}^{k}\frac{l_{i,\mathfrak{s}}^2\gamma_i^2}{2}\|y_i\|^2 + \Gamma_{i,k}\end{aligned} \tag{40}$$

where $\Gamma_{i,k} = \sum_{\mathfrak{s}=1}^{k} 1/2\|\varphi_{i,\mathfrak{s}}^*\|^2 + \sum_{\mathfrak{s}=1}^{k} 1/2l_{i,\mathfrak{s}}\lambda_{i,M}^2\ell_i^2$.

Step n: From (14), (24), (25), and (27), we have

$$\begin{aligned}\dot{H}_{i,n} &= \dot{\hat{x}}_{i,n} - \dot{\theta}_{i,nf} - \dot{\xi}_{i,n} - \dot{o}_{i,n} \\ &= \hat{\boldsymbol{\rho}}_i^T\mathbf{s}_i(\boldsymbol{v}_i) + l_{i,n}\left(\hat{\lambda}_i\bar{\bar{x}}_{i,1} - \hat{x}_{i,1}\right) + \hat{\varphi}_{i,n}^TQ_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) \\ &\quad + \hat{d}_{i,n} - \dot{\theta}_{i,nf} + \eta_{i,n}\xi_{i,n} - \hat{\boldsymbol{\rho}}_i^T\Delta\boldsymbol{u}_i - \dot{o}_{i,n}.\end{aligned} \tag{41}$$

Then, define

$$\begin{aligned}V_{i,n} = V_{i,n-1} &+ \frac{1}{2}H_{i,n}^2 + \frac{1}{2p_{i,n}}\tilde{\varphi}_{i,n}^T\tilde{\varphi}_{i,n} + \frac{1}{2r_{i,n}}\tilde{D}_i^T\tilde{D}_i \\ &+ \frac{1}{2\,\mathfrak{z}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i^T\tilde{\boldsymbol{\rho}}_i + \frac{1}{2\,\mathfrak{z}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^\star\tilde{\boldsymbol{\psi}}_i\end{aligned} \tag{42}$$

where $p_{i,n}$, $r_{i,1}$, $\mathfrak{z}_{i,\rho}$, and $\mathfrak{z}_{i,\psi}$ are positive constants. $\hat{D}_i$ represent the estimate of D_i, the definition of D_i will be elaborated upon later, satisfying $\hat{D}_i = D_i - \tilde{D}_i$. $\rho_i^\star = \mathrm{diag}(\rho_{i,1},\ldots,\rho_{i,m_i})$, $\hat{\boldsymbol{\psi}}_i$ represent the estimate of $\boldsymbol{\psi}_i$, $\hat{\boldsymbol{\psi}}_i = \boldsymbol{\psi}_i - \tilde{\boldsymbol{\psi}}_i$, $\boldsymbol{\psi}_i = [\psi_{i,1},\ldots,\psi_{i,m_i}]^T$ with

$$\psi_{i,a} = \begin{cases}\dfrac{1}{\rho_{i,a}}, & \rho_{i,a} > 0 \\ 0, & \rho_{i,a} = 0\end{cases} \tag{43}$$

for $a = 1,\ldots,m_i$.

Using (41) and (42), we have

$$\begin{aligned}\dot{V}_{i,n} = \dot{V}_{i,n-1} &+ H_{i,n}\left[\hat{\boldsymbol{\rho}}_i^T\boldsymbol{v}_i + l_{i,k}\left(\hat{\lambda}_i\bar{\bar{x}}_{i,1} - \hat{x}_{i,1}\right)\right. \\ &\quad + \varphi_{i,n}^{*T}Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) + \hat{d}_{i,n} - 2\bar{\varphi}_{i,n}^TQ_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) \\ &\quad \left.-\dot{\theta}_{i,nf} + \eta_{i,n}\xi_{i,n} - \dot{o}_{i,n}\right] - \frac{1}{r_{i,n}}\tilde{D}_i\dot{\hat{D}}_i \\ &+ \frac{1}{p_{i,n}}\tilde{\varphi}_{i,n}\left(p_{i,n}H_{i,n}Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) - \dot{\hat{\varphi}}_{i,n}\right) \\ &- \frac{1}{\mathfrak{z}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i\dot{\hat{\boldsymbol{\rho}}}_i - \frac{1}{\mathfrak{z}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^\star\dot{\hat{\boldsymbol{\psi}}}_i.\end{aligned} \tag{44}$$

From Lemma 5 and Young's inequality, we can get

$$\begin{aligned}\sum_{\mathfrak{s}=1}^{n}\sum_{\mathfrak{q}=1}^{\mathcal{N}} x_{\mathfrak{q},1}\hbar_{i\mathfrak{s}\mathfrak{q}}\left(\hat{x}_{i,1}\right) &\le \frac{nN}{2}\left\|\mathrm{e}_{i,1}\right\|^2 + \frac{1}{2}\sum_{\mathfrak{s}=1}^{n}\sum_{\mathfrak{q}=1}^{\mathcal{N}}\hbar_{i\mathfrak{s}\mathfrak{q}}^2\left(\hat{x}_{i,1}\right) \\ &+ \sum_{\mathfrak{s}=1}^{n}\sum_{\mathfrak{q}=1}^{\mathcal{N}}\hat{x}_{\mathfrak{q},1}\hbar_{i\mathfrak{s}\mathfrak{q}}\left(\hat{x}_{i,1}\right).\end{aligned} \tag{45}$$

$$\begin{aligned}&\frac{n}{2}\sum_{\mathfrak{q}=1}^{\mathcal{N}} x_{\mathfrak{q},1}^2 + \frac{\gamma_i^2}{2}\|y_i\|^2 + \sum_{\mathfrak{s}=1}^{n}\frac{l_{i,\mathfrak{s}}^2\gamma_i^2}{2}\|y_i\|^2 \\ &\le \left(nN + \gamma_i^2 + \sum_{\mathfrak{s}=1}^{n} l_{i,\mathfrak{s}}^2\gamma_i^2\right)\left(\left\|\mathrm{e}_{i,1}\right\|^2 + \left\|\hat{x}_{i,1}\right\|^2\right).\end{aligned} \tag{46}$$

Substituting (23), (45), and (46) into (44) gives

$$\begin{aligned}\dot{V}_{i,n} \le &-\mathrm{v}_i\left\|\mathbf{e}_i\right\|^2 - \sum_{\mathfrak{s}=1}^{n} a_{i,0}\tilde{W}_{i,\mathfrak{s}}^2 + 2\sum_{\mathfrak{s}=1}^{n}\left\|\tilde{\varphi}_{i,\mathfrak{s}}\right\|^2 + \Gamma_i \\ &+ \frac{1}{2}\left(\left\|\mathcal{P}_i\right\|^2\left\|\boldsymbol{\mathcal{L}}_i\right\|^2 + \sum_{\mathfrak{s}=1}^{n} l_{i,\mathfrak{s}}^2\right)\tilde{\lambda}_i^2\bar{\bar{x}}_{i,1}^2 + \left\|\tilde{\boldsymbol{\rho}}_i\right\|^2\left\|\check{\boldsymbol{\nu}}_i\right\|^2 \\ &+ \frac{\tilde{\lambda}_i^2}{q_{i,1}}\mathrm{sign}\left(\tilde{\lambda}_i\right)\left(\lambda_i - \dot{\hat{\lambda}}_i\right) + \sum_{\mathfrak{s}=1}^{n-1}\frac{\sigma_{i,\mathfrak{s}}}{p_{i,\mathfrak{s}}}\tilde{\varphi}_{i,\mathfrak{s}}^T\hat{\varphi}_{i,\mathfrak{s}} \\ &+ H_{i,n}\left(\hat{\boldsymbol{\rho}}_i^T\boldsymbol{v}_i + \hat{d}_{i,n} - \dot{\theta}_{i,nf} + \eta_{i,n}\xi_{i,n} + c_{i,n}o_{i,n}\right. \\ &\qquad + \frac{7}{2}o_{i,n} + 4\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right) \\ &\qquad \left.\frac{\hat{D}_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right)}{H_{i,n}} - \alpha_{i,n} + \alpha_{i,n}\right) \\ &+ \frac{7}{2}H_{i,n}^2 - \frac{1}{\mathfrak{z}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i\dot{\hat{\boldsymbol{\rho}}}_i - \frac{1}{\mathfrak{z}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^\star\dot{\hat{\boldsymbol{\psi}}}_i \\ &- \sum_{\mathfrak{s}=1}^{n-1} c_{i,\mathfrak{s}}H_{i,\mathfrak{s}}^2 + \frac{1}{p_{i,n}}\tilde{\varphi}_{i,n}^T\left(p_{i,n}H_{i,n}Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) - \dot{\hat{\varphi}}_{i,n}\right) \\ &+ \frac{1}{r_i}\tilde{D}_i\left(4r_i\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right) - \dot{\hat{D}}_i\right) \\ &+ 2\left(1 - 2\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\right)D_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right)\end{aligned} \tag{47}$$

where $\Gamma_{i,n} = \sum_{\mathfrak{s}=1}^{n} 1/2\|\varphi_{i,\mathfrak{s}}^*\|^2 + 1/3|(\bar{\lambda}_i)/(q_{i,1})|^3 + \sum_{\mathfrak{s}=1}^{n} 1/2l_{i,\mathfrak{s}}\lambda_{i,M}^2\ell_i^2$, $\mathrm{v}_i = \nu_{i,0} - \sum_{\mathfrak{s}=1}^{n}(l_{i,\mathfrak{s}}^2)/2 - (3nN)/(2) - \gamma_i^2 - \sum_{\mathfrak{s}=1}^{n} l_{i,\mathfrak{s}}^2\gamma_i^2$, $\Gamma_i = \Gamma_{i,n} + \Gamma_{i,0} + 1/2\bar{\partial}_i^2 + 3/4$, $D_i = \|\boldsymbol{G}^*{}_i\|^2$, and $\boldsymbol{G}^*{}_i^T\boldsymbol{S}_i(\hat{x}_{i,1})$ represents the approximations of unknown function $\mathfrak{M}$. ∂_i represents the approximation error and satisfies $|\partial_i| \le \bar{\partial}_i$, with $\bar{\partial}_i$ being a constant.

Based on (47), the virtual controller $\alpha_{i,n}$, the adaptive laws $\dot{\hat{\varphi}}_{i,n}$, and $\dot{\hat{D}}_i$ can be designed as

$$\begin{aligned}\alpha_{i,n} = &-c_{i,n}z_{i,n} - \frac{7}{2}z_{i,n} - \hat{d}_{i,n} + \dot{\theta}_{i,nf} - \eta_{i,n}\xi_{i,n} \\ &- 4\tanh^2\left(\frac{H_{i,n}}{\epsilon}\right)\frac{\hat{D}_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right)}{H_{i,n}}\end{aligned} \tag{48}$$

$$\dot{\hat{\varphi}}_{i,n} = p_{i,n}H_{i,n}Q_{i,n}\left(\underline{\hat{\boldsymbol{x}}}_{i,n}\right) - \sigma_{i,n}\hat{\varphi}_{i,n} \tag{49}$$

$$\dot{\hat{D}}_i = 4r_i\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right) - \varpi_i\hat{D}_i. \tag{50}$$

Substituting (48)–(50) into (47) gives

$$\dot{V}_{i,n} \le -\mathrm{v}_i\|\mathbf{e}_i\|^2 - \sum_{\mathfrak{s}=1}^{n} a_{i,0}\tilde{W}_{i,\mathfrak{s}}^2 + 2\sum_{\mathfrak{s}=1}^{n}\left\|\tilde{\varphi}_{i,\mathfrak{s}}\right\|^2$$

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$$+\frac{1}{2}\left(\|\mathcal{P}_i\|^2\left\|\|\mathcal{L}_i\|^2+\sum_{\mathfrak{s}=1}^{n}l_{i,\mathfrak{s}}^2\right)\tilde{\lambda}_i^2\bar{\bar{x}}_{i,1}^2+\Gamma_i\right.$$
$$+\frac{\tilde{\lambda}_i^2}{q_{i,1}}\text{sign}\left(\tilde{\lambda}_i\right)\left(\dot{\lambda}_i-\dot{\hat{\lambda}}_i\right)+\sum_{\mathfrak{s}=1}^{n}\frac{\sigma_{i,\mathfrak{s}}}{p_{i,\mathfrak{s}}}\tilde{\varphi}_{i,\mathfrak{s}}^T\hat{\varphi}_{i,\mathfrak{s}}$$
$$-\sum_{\mathfrak{s}=1}^{n}c_{i,\mathfrak{s}}H_{i,\mathfrak{s}}^2+\left\|\tilde{\boldsymbol{\rho}}_i\right\|^2\|\check{\boldsymbol{\nu}}_i\|^2-\frac{1}{\mathfrak{I}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^{\star}\dot{\hat{\boldsymbol{\psi}}}_i$$
$$-\frac{1}{\mathfrak{I}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i^T\dot{\hat{\boldsymbol{\rho}}}_i+H_{i,n}\left(\hat{\boldsymbol{\rho}}_i^T\boldsymbol{\nu}_i-\alpha_{i,n}\right)+\frac{\varpi_i}{r_i}\tilde{D}_i^T\hat{D}_i$$
$$+2\left(1-2\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\right)D_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right). \quad (51)$$

Then, the controller v_i, the adaptive laws $\dot{\hat{\boldsymbol{\rho}}}_i$, and $\dot{\hat{\boldsymbol{\psi}}}_i$ can be designed to

$$\boldsymbol{\nu}_i=-\hat{\boldsymbol{\psi}}_i\alpha_{i,n}\tanh\left(\frac{H_{i,n}\alpha_{i,n}}{\varepsilon}\right) \quad (52)$$
$$\dot{\hat{\boldsymbol{\rho}}}_i=\mathfrak{I}_{i,\rho}\,\hat{\boldsymbol{\psi}}_iH_{i,n}\alpha_{i,n}\tanh\left(\frac{H_{i,n}\alpha_{i,n}}{\varepsilon}\right)-\varkappa_{i,\rho}\hat{\boldsymbol{\rho}}_i \quad (53)$$
$$\dot{\hat{\boldsymbol{\psi}}}_i=I_{m_i}\,\mathfrak{I}_{i,\psi}\,H_{i,n}\alpha_{i,n}\tanh\left(\frac{H_{i,n}\alpha_{i,n}}{\varepsilon}\right)-\varkappa_{i,\psi}\hat{\boldsymbol{\psi}}_i \quad (54)$$

where $I_{m_i}=\text{diag}(1,1,\ldots,1)\in R^{m_i\times m_i}$.

According to Lemma 3 and (52)–(54), we can obtain

$$H_{i,n}\left(\hat{\boldsymbol{\rho}}_i^T\boldsymbol{\nu}_i-\alpha_{i,n}\right)-\frac{1}{\mathfrak{I}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i^T\dot{\hat{\boldsymbol{\rho}}}_i-\frac{1}{\mathfrak{I}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^{\star}\dot{\hat{\boldsymbol{\psi}}}_i$$
$$\le\frac{\varkappa_{i,\rho}}{\mathfrak{I}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i^T\hat{\boldsymbol{\rho}}_i+\frac{\varkappa_{i,\psi}}{\mathfrak{I}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^{\star}\hat{\boldsymbol{\psi}}_i+0.2785\varepsilon \quad (55)$$

where $\mathfrak{I}_i<m_i$ represents the total number of actuators in the ith subsystem that completely lose their effectiveness.

Substituting (52)–(55) into (51) gives

$$\dot{V}_{i,n}\le-\mathrm{v}_i\|\mathbf{e}_i\|^2-\sum_{\mathfrak{s}=1}^{n}a_{i,0}\tilde{W}_{i,\mathfrak{s}}^2+2\sum_{\mathfrak{s}=1}^{n}\left\|\tilde{\varphi}_{i,\mathfrak{s}}\right\|^2$$
$$-\left(\frac{b_i}{3}-\frac{2}{3}\right)|\tilde{\lambda}_i|^3+\frac{b_i}{3}\lambda_{i,M}^3+\frac{1}{3}\left|\frac{\bar{\tilde{\lambda}}_i}{q_{i,1}}\right|^3+\Gamma_i$$
$$-\sum_{\mathfrak{s}=1}^{n}c_{i,\mathfrak{s}}H_{i,\mathfrak{s}}^2+\sum_{\mathfrak{s}=1}^{n}\frac{\sigma_{i,\mathfrak{s}}}{p_{i,\mathfrak{s}}}\tilde{\varphi}_{i,\mathfrak{s}}^T\hat{\varphi}_{i,\mathfrak{s}}+\frac{\varpi_i}{r_i}\tilde{D}_i^T\hat{D}_i$$
$$+\frac{\varkappa_{i,\rho}}{\mathfrak{I}_{i,\rho}}\tilde{\boldsymbol{\rho}}_i^T\hat{\boldsymbol{\rho}}_i+\left\|\tilde{\boldsymbol{\rho}}_i\right\|^2\|\check{\boldsymbol{\nu}}_i\|^2+\frac{\varkappa_{i,\psi}}{\mathfrak{I}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^{\star}\hat{\boldsymbol{\psi}}_i$$
$$+2\left(1-2\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\right)D_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right)$$
$$+0.2785\varepsilon. \quad (56)$$

C. System Stability

Theorem 1: For the interconnected system (1) affected by output event-triggered (6), actuator faults (3), and input saturation (4), under Assumptions 1 and 4, through the design of the composite observer (14), the adaptive laws (34), (39), (49), (50), (53), and (54), the virtual controllers (33), (38), (48), and the control signal (52), it follows that all signals within the closed-loop system remain bounded. Additionally, the output signal $y_{i,1}$ can effectively follow the desired signal $y_{i,d}$, while also avoiding Zeno behavior.

Proof: The overall Lyapunov function is given by

$$V=\sum_{i=1}^{\mathcal{N}}V_{i,n}. \quad (57)$$

Based on Young's inequality and (56), we are able to acquire

$$\dot{V}\le\sum_{i=1}^{\mathcal{N}}\left(-\mathrm{v}_i\|\mathbf{e}_i\|^2-\sum_{\mathfrak{s}=1}^{n}a_{i,0}\tilde{W}_{i,\mathfrak{s}}^2-\left(\frac{b_i}{3}-\frac{2}{3}\right)|\tilde{\lambda}_i|^3\right.$$
$$-\sum_{\mathfrak{s}=1}^{n}c_{i,\mathfrak{s}}H_{i,\mathfrak{s}}^2-\sum_{\mathfrak{s}=1}^{n}\left(\frac{\sigma_{i,\mathfrak{s}}}{2p_{i,\mathfrak{s}}}-2\right)\tilde{\varphi}_{i,\mathfrak{s}}^T\tilde{\varphi}_{i,\mathfrak{s}}+\Gamma_i$$
$$-\frac{\varpi_i}{2r_i}\tilde{D}_i^T\tilde{D}_i-\left(\frac{\varkappa_{i,\rho}}{2\,\mathfrak{I}_{i,\rho}}-\|\check{\boldsymbol{\nu}}_i\|^2\right)\tilde{\boldsymbol{\rho}}_i^T\tilde{\boldsymbol{\rho}}_i+\frac{1}{3}\left|\frac{\bar{\tilde{\lambda}}_i}{q_{i,1}}\right|^3$$
$$-\frac{\varkappa_{i,\psi}}{2\,\mathfrak{I}_{i,\psi}}\tilde{\boldsymbol{\psi}}_i^T\rho_i^{\star}\tilde{\boldsymbol{\psi}}_i+\frac{\sigma_{i,s}}{2p_{i,s}}\left\|\varphi_{i,s}\right\|^2+\frac{\varkappa_{i,\rho}}{2\,\mathfrak{I}_{i,\rho}}\left\|\boldsymbol{\rho}_i\right\|^2$$
$$+\frac{\varpi_i}{2r_i}\|D_i\|^2+\frac{\varkappa_{i,\psi}}{2\,\mathfrak{I}_{i,\psi}}\boldsymbol{\psi}_i^T\rho_i^{\star}\boldsymbol{\psi}_i$$
$$\left.+0.2785\varepsilon+\frac{b_i}{3}\lambda_{i,M}^3\right)\le-gV+\iota+\mathfrak{f} \quad (58)$$

where $g=\min\{(\mathrm{v}_i)/(\lambda_{\max}\{\mathcal{P}_i\}),2a_{i,0},b_i-2,2c_{i,s},(\sigma_{i,s})/(p_{i,s})-4,(\varpi_i)/(r_i),(\varkappa_{i,\rho})/(\mathfrak{I}_{i,\rho})-2\|\check{\boldsymbol{\nu}}_i\|^2,(\varkappa_{i,\psi})/(\mathfrak{I}_{i,\psi})\}$, $\iota=(b_i)/3\lambda_{i,M}^3+1/3|(\bar{\tilde{\lambda}}_i)/(q_{i,1})|^3+\Gamma_i+(\sigma_{i,s})/(2p_{i,s})\|\varphi_{i,s}\|^2+(\varkappa_{i,\rho})/(2\;\mathfrak{I}_{i,\rho})\|\boldsymbol{\rho}_i\|^2+(\varpi_i)/(2r_i)\|D_i\|^2+(\varkappa_{i,\psi})/(2\;\mathfrak{I}_{i,\psi})\boldsymbol{\psi}_i^T\rho_i^{\star}\boldsymbol{\psi}_i+0.2785\varepsilon$, $\mathfrak{f}=2(1-2\tanh^2((H_{i,n})/(\epsilon_i)))D_i\boldsymbol{S}_i^T(\hat{x}_{i,1})\boldsymbol{S}_i(\hat{x}_{i,1})$.

In light of Lemma 4, the following two cases should be taken into account.

Case 1: When $H_{i,1}\notin\Omega_\epsilon=\{H_{i,1}\|H_{i,1}|\le0.8814\epsilon\}$, we get

$$\mathfrak{f}=2\left(1-2\tanh^2\left(\frac{H_{i,n}}{\epsilon_i}\right)\right)D_i\boldsymbol{S}_i^T\left(\hat{x}_{i,1}\right)\boldsymbol{S}_i\left(\hat{x}_{i,1}\right)\le0. \quad (59)$$

Therefore, based on (58), we can obtain

$$\dot{V}\le-gV+\iota. \quad (60)$$

Then, one has

$$0\leqslant V(t)\le V(0)e^{-gt}+\frac{\iota}{g}\left(1-e^{-gt}\right). \quad (61)$$

From (61), we have that $H_{i,k}$, $z_{i,k}$, $\mathbf{e}_i$, $\hat{\varphi}_{i,k}$, $\hat{\boldsymbol{\rho}}_i$, $\hat{\boldsymbol{\psi}}_i$, $\hat{D}_i$ and $\hat{\lambda}_i$ are bounded as $t\to\infty$. To verify the boundedness of the auxiliary variables $\xi_{i,k}$, a Lyapunov function $V_\xi=1/2\sum_{i=1}^{\mathcal{N}}\sum_{k=1}^{n}\xi_{i,k}^2$ is constructed. According to (24), one can readily derive $\dot{V}_\xi\le-\eta_0V_\xi+1/2(\hat{\boldsymbol{\rho}}_i^T\Delta\boldsymbol{u}_i)^2$, where $\eta_0=\min_{1\le i\le\mathcal{N},1\le k\le n}\bar{\eta}_{i,k}$ with $\bar{\eta}_{i,1}=\eta_{i,1}-1/2$ and $\bar{\eta}_{i,\mathfrak{b}}=\eta_{i,\mathfrak{b}}-1(\mathfrak{b}=2,\ldots,n)$, from which it follows that $V_\xi\le1/(2\eta_0)(\hat{\boldsymbol{\rho}}_i^T\Delta\boldsymbol{u}_i)^2$, implying that $\xi_{i,k}$ is bounded. Furthermore, based on Assumption 1 and (25), it can be derived that all other control signals are bounded. Let $\eta_i=x_{i,1}-y_{i,d}$, we can get $\|\eta_i\|\le(2\iota/g)^{1/2}+\|\mathrm{e}_{i,1}\|+\|\xi_{i,1}\|+(\mu_i)/2\mathfrak{r}_i$, which means that the tracking error converges to a bounded compact set.

Case 2: When $H_{i,1}\in\Omega_\epsilon=\{H_{i,1}\|H_{i,1}|\le0.8814\epsilon\}$, we can conclude that $H_{i,1}$ is bounded. As D_i is bounded,

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$\mathfrak{f}$ is also bounded, and assuming that the upper bound of $\mathfrak{f}$ is $\mathfrak{f}_M$. Then, (58) becomes

$$\dot{V} \leq -gV + \iota + \mathfrak{f}_M. \tag{62}$$

In a manner similar to *Case 1*, one can deduce that all signals within the closed-loop system are bounded. Then, we can obtain $\|\eta_i\| \leq (2(\iota + \mathfrak{f}_M)/(g))^{1/2} + \|\mathrm{e}_{i,1}\| + \|\xi_{i,1}\| + (\mu_i)/(2\mathfrak{r}_i)$, which means that the tracking error converges to a bounded compact set.

Finally, we verify that Zeno's behavior is eliminated. Define $E_{i,\mathfrak{p}}(t) = \check{y}_i(t) - \bar{\bar{y}}_i(t)\ \forall t \in [t_{i,\mathfrak{p}}, t_{i,\mathfrak{p}+1})$, $i = 1, \ldots, \mathcal{N}$, whose derivatives are computed as

$$\frac{d}{dt}\left|E_{i,\mathfrak{p}}\right| = \frac{d}{dt}\left(E_{i,\mathfrak{p}} \times E_{i,\mathfrak{p}}\right)^{\frac{1}{2}} = \operatorname{sign}\left(E_{i,\mathfrak{p}}\right)\dot{E}_{i,\mathfrak{p}} \leq \left|\dot{\check{y}}_i\right|. \tag{63}$$

From Theorem 1, we obtain that $x_{i,k}$ and λ_i are all bounded. So, there must be constant $\mathbb{k}_i > 0$ that satisfies $|\dot{\check{y}}_i| \leq \mathbb{k}_i$. Let $\Delta t = t_{i,\mathfrak{p}+1} - t_{i,\mathfrak{p}}$, we get $\lim_{\Delta t \to 0} |(E_{i,\mathfrak{p}+1}(t_{,\mathfrak{p}+1}) - E_{i,\mathfrak{p}}(t_{i,\mathfrak{p}}))/(\Delta t)| = |\dot{E}_{i,\mathfrak{p}}|$. Since $E_{i,\mathfrak{p}}(t_{i,\mathfrak{p}}) = 0$, it can be concluded that $\Delta t \geq (\gamma_i|\check{y}_i| + \ell_i)/(\mathbb{k}_i)$. This implies that Zeno's behavior was avoided. ■

IV. Simulation Studies

In this section, numerical and comparative simulations are presented to validate the effectiveness of the proposed decentralized resilient adaptive ETC strategy.

A. Numerical Simulation

Consider the following interconnected system composed of two subsystems whose dynamic model can be described as

$$\begin{cases} \dot{x}_{1,1} = x_{1,2} - 5.3\sin\left(x_{1,1}\right) + 0.01\sin\left(x_{1,1}\right)x_{2,1}^2 + d_{1,1} \\ \dot{x}_{1,2} = \sum_{\mathfrak{a}=1}^{2} u_{1,\mathfrak{a}} + 13x_{1,1}\sin\left(x_{1,2} + 0.05\left(\sin\left(x_{1,1}\right)\right.\right. \\ \qquad \left.\left. - \sin\left(x_{2,1}\right)\right) + d_{2,1} \\ y_1 = x_{1,1} \\ x_{2,1} = x_{2,2} - 6\sin\left(x_{2,1}\right) + 0.01\sin\left(x_{2,1}\right)x_{1,1}^2 + d_{2,1} \\ \dot{x}_{2,2} = \sum_{\mathfrak{a}=1}^{2} u_{2,\mathfrak{a}} + 5.1x_{2,1}\sin\left(x_{2,2}\right) + 0.1\left(\sin\left(x_{1,1}\right)\right. \\ \qquad \left. - \sin\left(x_{2,1}\right)\right) + d_{2,2} \\ y_2 = x_{2,1}. \end{cases} \tag{64}$$

Select the saturation threshold as $\bar{v}_{1,1} = 1.3$, $\underline{v}_{1,1} = -1.3$, $\bar{v}_{2,1} = 1$, and $\underline{v}_{2,2} = 1$. The actuator fault model is shown below

$$u_{1,1} = \begin{cases} 0.4\mathrm{s}_{1,1}\left(v_{1,1}\right), & t \geq 30s \\ \mathrm{s}_{1,1}\left(v_{1,1}\right), & \text{otherwise} \end{cases} \tag{65}$$

$$u_{1,2} = \begin{cases} 0.3\mathrm{s}_{1,2}\left(v_{1,2}\right), & t \geq 45s \\ \mathrm{s}_{1,2}\left(v_{1,2}\right), & \text{otherwise} \end{cases} \tag{66}$$

$$u_{2,1} = \begin{cases} 0.2\mathrm{s}_{2,1}\left(v_{2,1}\right), & t \geq 55s \\ \mathrm{s}_{2,1}\left(v_{2,1}\right), & \text{otherwise} \end{cases} \tag{67}$$

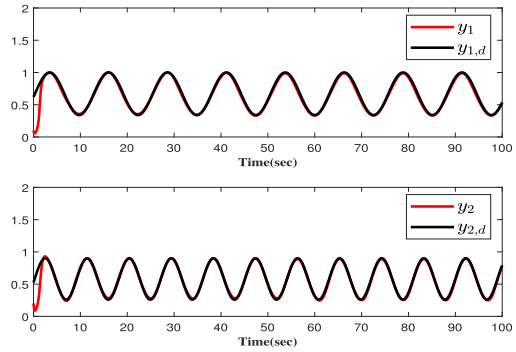


Fig. 2. Trajectories of y_i and $y_{i,d}$, $i = 1, 2$.

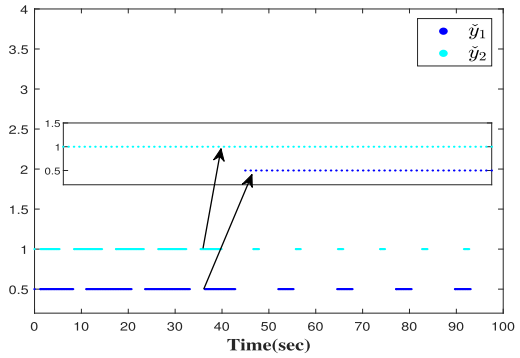


Fig. 3. Triggering instants of $\check{y}_i$, $i = 1, 2$.

and

$$u_{2,2} = \mathrm{s}_{2,2}(v_{2,2}). \tag{68}$$

The deception attack is shown below

$$\lambda_i = \begin{cases} -e^{-t} + 2, & t \geq 40s \\ 1, & \text{otherwise.} \end{cases} \tag{69}$$

In the simulation, the designed desired trajectories $y_{1,d} = 0.33\sin(0.5t - 0.15) + 0.66$ and $y_{2,d} = 0.32\sin(0.7t - 0.175) + 0.58$, and the external disturbance $d_{1,1} = d_{1,2} = 1.2\sin(0.5t)$ and $d_{2,1} = d_{2,2} = 1.2\sin(0.7t)$. Choosing $x_1(0) = [0.1, 0.1]^T$, $x_2(0) = [0.2, 0.2]^T$ as the initial states, the triggering condition parameters are designed to $\gamma_1 = \gamma_2 = 0.4$ and $\ell_1 = 0.26$, $\ell_2 = 0.2$, the parameters are as follows: $c_{11} = c_{21} = 1$, $c_{12} = c_{22} = 0.5$, $l_{1,1} = l_{21} = -0.1$, $l_{12} = l_{2,2} = -0.1\tau_{1,2} = \tau_{2,2} = 10$, $\pi_{1,1} = \pi_{2,1} = 1$, $\pi_{1,2} = \pi_{22} = 0.1$, $q_{1,1} = q_{1,2} = 1$, $b_1 = 0.05$, $b_2 = 0.02$, $p_{1,1} = p_{1,2} = 10$, $p_{1,2} = p_{2,2} = 10$, $\sigma_{1,1} = \sigma_{2,1} = \sigma_{2,2} = 2$, $\sigma_{1,2} = 5$, $\eta_{1,1} = \eta_{1,2} = \eta_{2,2} = 2$, $\eta_{2,1} = 1$, $r_1 = r_2 = 5$, $\epsilon_1 = \epsilon_2 = 3$, $\varpi_1 = \varpi_2 = 2$, $\mathfrak{z}_{1,\rho} = 3$, $\mathfrak{z}_{2,\rho} = 0.5$, $\mathfrak{z}_{1,\psi} = 2$, $\mathfrak{z}_{2,\psi} = 1$, $\varepsilon = 1$, $\varkappa_{1,\rho} = \varkappa_{2,\rho} = 0.5$, AND $\varkappa_{1,\psi} = \varkappa_{2,\psi} = 0.1$.

Simulation results are presented in Figs. 2–5, respectively. Fig. 2 represents the trajectories of output signal y_i and desired signal $y_{i,d}$. Fig. 3 shows the triggering instants of $\check{y}_i$. Fig. 4 displays the trajectories of the actual controller $u_{i,s}$ and the designed controller $v_{i,s}$. Fig. 5 displays the trajectories of the actuator fault compensation signals $\hat{\rho}_i$ and $\hat{\psi}_i$.

B. Comparative Simulation

To verify the effectiveness and superiority of the proposed control strategy, we compared it with the traditional control

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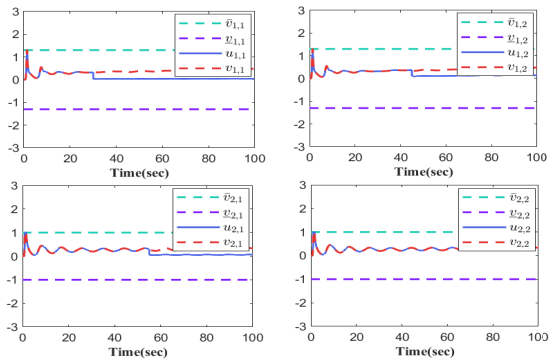


Fig. 4. Trajectories of the actual controller $u_{i,s}$ and the designed controller $v_{i,s}$, $i = 1, 2$, $s = 1, 2$.

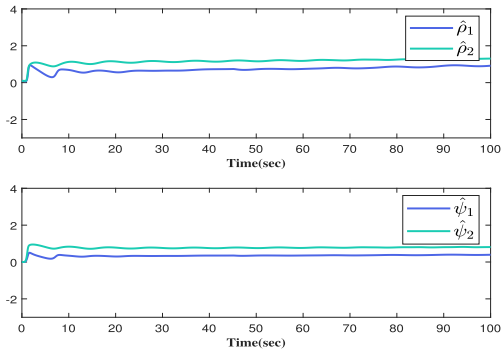


Fig. 5. Trajectories of the actuator fault compensation signals $\hat{\boldsymbol{\rho}}_i$ and $\hat{\boldsymbol{\psi}}_i$, $i = 1, 2$.

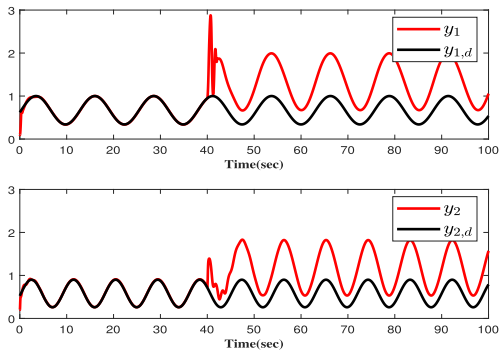


Fig. 6. Trajectories of y_i and $y_{i,d}$ by the method in [38], $i = 1, 2$.

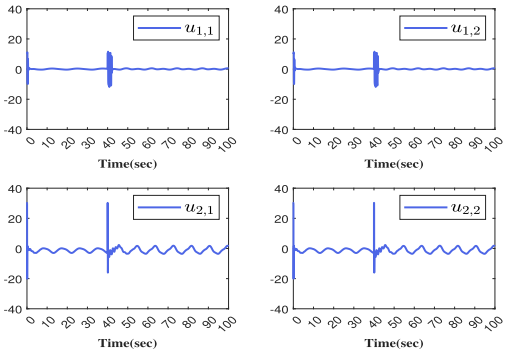


Fig. 7. Trajectories of inputs $u_{i,a}$ by the method in [38], $i = 1, 2$, $a = 1, 2$.

strategy in [38], which does not incorporate attack compensation. The comparative results are shown in Figs. 6–8. The

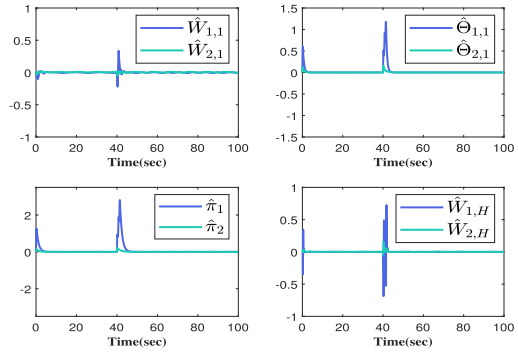


Fig. 8. Trajectories of the parameter adaptive laws by the method in [38].

simulation results indicate that when deception attacks occur at $t = 40$ s, the system output trajectories deviate noticeably from the desired trajectories shown in Fig. 6, while significant fluctuations appear in the control inputs and adaptive parameter trajectories shown in Figs. 7 and 8. Therefore, the control strategy proposed in this article effectively saves communication and computational resources while ensuring tracking performance and enhancing the practicality of the control system.

V. Conclusion

This article investigates the decentralized adaptive intermittent output feedback tracking problem for a class of nonlinear interconnected systems subject to multiple actuator faults, input saturation, and deception attacks. A new composite observer is constructed using intermittent compromised output signals and FLSs to estimate unknown state variables and handle external disturbances. A projection operator-based compensation mechanism is proposed to handle unknown gains caused by deception attacks. Based on Lyapunov stability analysis, all closed-loop system signals are bounded, and the tracking errors converge to a bounded compact set. In future studies, the method presented in this article will be extended to multiagent systems, taking into account performance constraints and resource optimization, so as to further enhance its practical applicability.

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